

NanoGT Disease Mechanisms — Curated Molecular Evidence Table

Source: data/disease_mechanisms.csv | Project: Bridging the Nano-Rare Gap | 30-disease cohort

COMPATIBILITY LEGEND									
Compatible — Mechanism directly supports gene addition			Conditional — Supports gene addition with caveats			Uncertain — Not standard gene addition; specialist strategy needed			
Disease Name	ORPHA ID	Gene	Mechanism Category	Mechanism Detail	Gene-Additi on Compati bility	Preferred Modality	Evidence Summary	Evidence Citation	Evidence Status
Hemophilia B	ORPHA:306	F9	Loss Of Function	Factor IX deficiency from pathogenic F9 variants	Compatible	Gene addition / factor expression	Deficiency of coagulation factor IX is the therapeutic deficit; liver-directed F9 expression is an approved gene-addition precedent	OMIM Hemophilia B 306900	Source linked — needs review
Leber congenital amaurosis	ORPHA:65	RPE65	Loss Of Function	RPE65 retinoid-cycle enzyme deficiency	Compatible	Gene addition	Biallelic RPE65 disease is treated by supplying functional RPE65 to retinal pigment epithelium	FDA Luxturna product information	Source linked — needs review
Severe combined immunodeficiency due to adenosine deaminase deficiency	ORPHA:277	ADA	Loss Of Function	ADA enzyme deficiency causing toxic purine metabolite accumulation	Compatible	Ex vivo HSC gene addition	ADA-SCID is a deficiency disorder with approved ex vivo gene-addition precedent	OMIM ADA-SCID 102700	Source linked — needs review
Spinal Muscular Atrophy	ORPHA:70	SMN1	Loss Of Function	SMN protein deficiency from biallelic SMN1 mutation	Compatible	Gene addition	FDA Zolgensma indication is SMA with biallelic SMN1 mutations; product supplies functional SMN transgene	FDA Zolgensma product page	Source linked — needs review
Duchenne muscular dystrophy	ORPHA:98896	DMD	Loss Of Function Oversized	Dystrophin loss with native DMD too large for single AAV	Conditional	Engineered microgene / dual vector	DMD supports replacement logic but requires micro-dystrophin or other engineered strategies because full-length DMD exceeds AAV capacity	Chamberlain et al. 2023 microdystrophin review	Source linked — needs review
Leber hereditary optic neuropathy	ORPHA:104	MT-ND4	Mitochondrial Loss Of Function	Mitochondrial complex I dysfunction from MT-ND4 variant	Uncertain	Allotopic / mitochondrial strategy	Mitochondrial DNA disease is not ordinary nuclear gene addition; allotopic AAV approaches are specialized and disease-specific	OMIM MT-ND4 gene entry	Needs manual review
Achromatopsia	ORPHA:49382	CNGB3	Loss Of Function	Cone cyclic nucleotide-gated channel beta-subunit loss	Compatible	Retinal gene addition	Biallelic CNGB3 loss is compatible with photoreceptor-directed gene addition if target cells remain viable	OMIM CNGB3 gene entry	Source linked — needs review
Alpha-mannosidosis	ORPHA:61	MAN2B1	Loss Of Function	Lysosomal alpha-mannosidase enzyme deficiency	Compatible	Gene addition / cross-correction	Lysosomal enzyme deficiency supports gene addition and possible cross-correction logic	OMIM MAN2B1 gene entry	Source linked — needs review

Disease Name	ORPHA ID	Gene	Mechanism Category	Mechanism Detail	Gene-Addition Compatibility	Preferred Modality	Evidence Summary	Evidence Citation	Evidence Status
Choroideremia	ORPHA:180	CHM	Loss Of Function	REP1 deficiency affecting retinal cells	Compatible	Retinal gene addition	CHM loss of function is a retinal gene-addition target if viable retina remains	OMIM CHM gene entry	Source linked — needs review
Crigler-Najjar syndrome type I	ORPHA:1060	UGT1A1	Loss Of Function	Absent or severe deficiency of bilirubin UDP-glucuronosyltransferase	Compatible	Liver gene addition	UGT1A1 deficiency supports hepatocyte-directed gene addition	OMIM UGT1A1 gene entry	Source linked — needs review
Fabry disease	ORPHA:324	GLA	Loss Of Function	Alpha-galactosidase A lysosomal enzyme deficiency	Compatible	Gene addition / cross-correction	Fabry is caused by deficient GLA enzyme activity; secreted enzyme strategies can support systemic cross-correction	OMIM GLA gene entry	Source linked — needs review
Gaucher disease	ORPHA:355	GBA	Loss Of Function	Beta-glucocerebrosidase lysosomal enzyme deficiency	Conditional	Gene addition / cross-correction	GBA deficiency supports replacement logic but neuronopathic disease and tissue access can limit simple systemic correction	OMIM GBA gene entry	Source linked — needs review
Glycogen storage disease type Ia	ORPHA:79258	G6PC	Loss Of Function	Glucose-6-phosphatase catalytic subunit deficiency	Compatible	Liver + kidney gene addition	G6PC deficiency is a metabolic enzyme loss where liver-directed expression is a plausible gene-addition strategy	OMIM G6PC gene entry	Source linked — needs review
Kohlschutter-Tonz syndrome	ORPHA:1946	ROGDI	Loss Of Function	ROGDI loss of function from recessive truncating splice or destabilizing variants	Conditional	CNS gene addition (cell-autonomous)	Published KTS variants in ROGDI are consistent with loss of function; gene addition is plausible but ROGDI is intracellular and disease biology is not fully resolved	Schossig et al. 2012 Mutations in ROGDI Cause Kohlschutter-Tonz Syndrome	Source checked
Krabbe disease	ORPHA:487	GALC	Loss Of Function	Galactocerebrosidase lysosomal enzyme deficiency	Conditional	Gene addition / HSC cross-correction	GALC deficiency supports replacement logic but CNS delivery and toxic metabolite biology require disease-specific validation	OMIM GALC gene entry	Source linked — needs review
Maple syrup urine disease	ORPHA:511	BCKDHA	Loss Of Function	Branched-chain alpha-ketoacid dehydrogenase E1 alpha deficiency	Conditional	Liver gene addition / enzyme complex	BCKDHA deficiency supports replacement logic but the multi-subunit enzyme complex and CNS metabolic crises make liver-only rescue uncertain	OMIM BCKDHA gene entry	Source linked — needs review
Metachromatic leukodystrophy	ORPHA:512	ARSA	Loss Of Function	Arylsulfatase A lysosomal enzyme deficiency	Compatible	Ex vivo HSC / CNS gene addition	ARSA deficiency is compatible with gene addition and HSC-mediated enzyme delivery/cross-correction precedent	OMIM ARSA gene entry	Source linked — needs review

Disease Name	ORPHA ID	Gene	Mechanism Category	Mechanism Detail	Gene-Addition Compatibility	Preferred Modality	Evidence Summary	Evidence Citation	Evidence Status
Mucopolidosis type IV	ORPHA:578	MCOLN1	Loss Of Function	TRPML1 lysosomal membrane channel deficiency	Conditional	Cell-autonomous gene addition	MCOLN1 loss supports gene addition but membrane-channel biology is cell-autonomous and not a strong cross-correction case	OMIM MCOLN1 gene entry	Source linked — needs review
Mucopolysaccharidosis type I	ORPHA:579	IDUA	Loss Of Function	Alpha-L-iduronidase lysosomal enzyme deficiency	Compatible	Gene addition / cross-correction	IDUA enzyme deficiency supports replacement and cross-correction biology	OMIM MPS I 252800	Source linked — needs review
Mucopolysaccharidosis type II	ORPHA:580	IDS	Loss Of Function	Iduronate-2-sulfatase lysosomal enzyme deficiency	Compatible	Gene addition / cross-correction	IDS enzyme deficiency supports replacement and cross-correction biology	OMIM IDS gene entry	Source linked — needs review
Mucopolysaccharidosis type IIIA (Sanfilippo A)	ORPHA:79269	SGSH	Loss Of Function	Heparan N-sulfatase lysosomal enzyme deficiency	Compatible	CNS gene addition / cross-correction	SGSH enzyme deficiency supports replacement logic but CNS delivery is central for Sanfilippo A	OMIM SGSH gene entry	Source linked — needs review
Ornithine transcarbamylase deficiency	ORPHA:664	OTC	Loss Of Function	Ornithine transcarbamylase urea-cycle enzyme deficiency	Compatible	Liver gene addition	OTC deficiency is a liver metabolic enzyme loss compatible with hepatocyte gene addition	OMIM OTC gene entry	Source linked — needs review
Phenylketonuria	ORPHA:716	PAH	Loss Of Function	Phenylalanine hydroxylase enzyme deficiency	Compatible	Liver gene addition	PAH deficiency is a liver enzyme loss compatible with hepatocyte gene addition	OMIM PAH gene entry	Source linked — needs review
Pompe disease	ORPHA:365	GAA	Loss Of Function	Acid alpha-glucosidase lysosomal enzyme deficiency	Conditional	Gene addition / secreted enzyme	GAA deficiency supports replacement logic but skeletal and cardiac delivery needs broad tissue correction	OMIM GAA gene entry	Source linked — needs review
Salla disease	ORPHA:309	SLC17A5	Loss Of Function	Sialin lysosomal transporter deficiency	Conditional	Cell-autonomous gene addition	SLC17A5 transporter deficiency supports replacement logic but membrane transporter biology is mostly cell-autonomous	OMIM SLC17A5 gene entry	Source linked — needs review
Vitamin B12-unresponsive methylmalonic acidemia	ORPHA:27	MUT	Loss Of Function	Methylmalonyl-CoA mutase mitochondrial enzyme deficiency	Conditional	Liver gene addition / mitochondrial	MUT deficiency supports replacement logic but mitochondrial enzyme import and multi-system metabolic disease require validation	OMIM MUT gene entry	Source linked — needs review
Wiskott-Aldrich syndrome	ORPHA:906	WAS	Loss Of Function	WASP deficiency in hematopoietic cells	Compatible	Ex vivo HSC gene addition	WAS loss of function is compatible with autologous hematopoietic stem-cell gene addition	OMIM WAS gene entry	Source linked — needs review
X-linked adrenoleukodystrophy	ORPHA:43	ABCD1	Loss Of Function	Peroxisomal ABCD1 transporter deficiency	Conditional	Ex vivo HSC gene addition	ABCD1 deficiency supports replacement logic but cerebral inflammatory disease relies on hematopoietic and CNS disease biology	OMIM ABCD1 gene entry	Source linked — needs review

Disease Name	ORPHA ID	Gene	Mechanism Category	Mechanism Detail	Gene-Addition Compatibility	Preferred Modality	Evidence Summary	Evidence Citation	Evidence Status
X-linked myotubular myopathy	ORPHA:596	MTM1	Loss Of Function	Myotubularin deficiency in skeletal muscle	Compatible	Muscle gene addition	MTM1 loss of function is compatible with muscle-directed gene addition but dose toxicity must be reviewed	OMIM MTM1 gene entry	Source linked — needs review
X-linked retinoschisis	ORPHA:792	RS1	Loss Of Function	Retinoschisin deficiency affecting retinal architecture	Compatible	Retinal gene addition	RS1 deficiency supports retinal gene addition if target retinal structure remains treatable	OMIM RS1 gene entry	Source linked — needs review

Note: Evidence status "Source linked — needs review" indicates a source URL has been recorded but requires final verification against the cited resource. "Source checked" indicates the claim has been verified. "Needs manual review" indicates the evidence requires expert assessment before use. Mechanism detail and compatibility labels are based on published literature and should be confirmed with a supervisor before submission.